



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester M.C.A. (CBCS) Degree Examinations
 January 2024

22MCA344 – NETWORK AND CYBER SECURITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

		Marks	BT*	CO*	PO*
Unit – I					
1. a)	Use the Caesar cipher to encrypt and decrypt the message "HELLO" with the key (shift) value 15.	10	L*3	1	1
b)	Apply polyalphabetic cipher for the text "JAVATPOINT". Consider the key as "BEST".	10	L3	1	1
2. a)	Apply Hill Cipher algorithm to encrypt the word "TEXT" and decrypt it using the 2x2 key matrix given below. $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$	10	L3	1	1
b)	Differentiate between active & passive attacks. Classify the following into active and passive attacks and justify the same. i) Masquerade, ii) Replay, iii) Evesdropping, iv) Traffic Analysis.	10	L2	1	1
Unit – II					
3. a)	In an RSA cryptosystem, a particular user A uses two prime numbers, 13 and 17, to generate the public and private keys. If the public key of A is 35, compute the private key of A.	10	L3	2	1
b)	Explain Millar-Rabin algorithm.	10	L2	2	1
4. a)	Suppose that two parties A and B wish to set up a common secret key (D-H key) between themselves using the Diffie Hellman key exchange technique. They agree on 7 as the modulus and 3 as the primitive root. Party A chooses 2 and party B chooses 5 as their respective secrets. Compute the shared secret key.	10	L3	2	1
b)	Explain Hash functions based on cipher block chaining with a neat diagram.	10	L2	2	1
Unit – III					
5. a)	Explain the general X. 509 public key certificate format.	10	L2	3	1
b)	Explain the Kerberos version 4 message exchanges.	10	L2	3	1
a)	List out the differences between Kerberos version 4 and version 5.	10	L2	3	1
b)	Explain the various 509 public key certificate Authentication Procedures.	10	L2	3	1
Unit – IV					
7. a)	Explain the classification of cybercrimes by giving examples for each.	10	L1	4	1
b)	Explain any five cyber laws under IPC.	10	L1	4	1
8. a)	Explain cybercafe and cyberstalking with relevant examples.	10	L2	4	1
b)	Explain botnets and the different types of attacks that could be performed using botnets.	10	L2	4	1
Unit – V					
9. a)	Explain DoS and DDoS attacks.	10	L2	5	1
b)	Explain SQL Injection to database and Buffer overflow attacks.	10	L2	5	1
10. a)	Explain any five attacks on Wireless networks.	10	L1	5	1
b)	Explain the following: i) Phishing, ii) Virus and Worms, iii) Trojan Horses and Backdoors	10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
